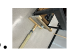
Data Review:

Our design underwent testing for adjustability, user comfort, stability, and compactness. Through physical tests and user feedback, we analyzed how effectively the design helps minimize back strain and enhances user comfort. While most of the data collected was qualitative, we made iterative adjustments to the product based on this feedback. The overall analysis indicates that our design successfully addresses the issue of back strain for cart operators.

After testing, assembling the product typically takes about 1.5 minutes, often requiring two people due to the need for attaching clamps to the cart. Although the assembly process is slightly lengthened by the use of clamps, these components are essential for providing perfect stability.

As shown in photo below:

One noted drawback is a slight reduction in control compared to the original cart, attributed to the handle's greater distance from the cart. Mr.Kiser claimed that even though the handle was in a comfortable spot to grab, it was awkwardly far from the cart. Despite this, the ergonomic benefits and enhanced stability make it a valuable improvement for cart operators. 

Mr.Kiser stated that the handle was not only comfortable to hold but was more ergonomic than the original cart handles.

The data suggest that our design is a significant improvement in terms of ergonomics and user comfort, making it a viable solution for reducing back strain among cart operators. The need for two-person assembly and the slight loss of control are minor trade-offs compared to the benefits of enhanced stability and adjustability.